

Data wrangling with Pandas: statistics and re-shaping

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After seeing how to associate datasets through joins, this chapter continues exploring the data scientist's toolbox with two classic operations: group descriptive statistics and data reshaping. It also introduces building communication-ready tables with `great_tables`.

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Skills to be acquired by the end of this chapter

- How to construct fine aggregate statistics using `Pandas` methods;
- Know how to reshape your data from *long* to *wide* format (and vice versa);
- Create attractive tables to communicate aggregated results.

1 Introduction

The [previous chapter](#) showed how to enrich a dataset by associating it with another source through a join. We now have two datasets that have been made consistent with one another: the Ademe greenhouse gas emissions data (`emissions`) and the Insee Filosofi framing data (`filosofi`).

This chapter will use this association to deepen our understanding of the data through two complementary operations:

- group descriptive statistics;
- reshaping data between *long* and *wide* formats.

Finally, we will see how to build, from these statistics, polished communication tables using the `great_tables` package.

1.1 Data

We start again from the same datasets as in the [chapter on joins](#): the Ademe municipal emissions data (`emissions`) and the Insee Filosofi data (`filosofi`), enriched with one another. The code below, already explained in detail in the previous chapter, retrieves these datasets again:

```
import requests
import pandas as pd

def get_melodi(dataset, dimension, code, value_name, time_period, geo_level="COM"):

    url = f"https://api.insee.fr/melodi/data/{dataset}"

    params = {
        "GEO": geo_level,
        "dimension": code,
        "TIME_PERIOD": time_period,
        "maxResult": 40000,
    }

    response = requests.get(url, params=params, timeout=60)
    response.raise_for_status()
    observations = response.json()["observations"]

    return pd.DataFrame(
        {
            "CODGEO": obs["dimensions"]["GEO"].split("-")[-1],
            value_name: obs["measures"]["OBS_VALUE_NIVEAU"].get("value"),
        }
        for obs in observations
    )

niveau_vie = get_melodi("DS_FILOSOFI_CC", "FILOSOFI_MEASURE", "MED_SL", "NIVVIE_MEDIAN", 2023)
taux_pauvrete = get_melodi("DS_FILOSOFI_CC", "FILOSOFI_MEASURE", "PR_MD60", "TAUX_PAUVRETE",
2023)
population = get_melodi("DS_POPULATIONS_HISTORIQUES", "POPREF_MEASURE", "PMUN", "POPULATION",
2023)

url_cog_2023 = "https://www.insee.fr/fr/statistiques/fichier/6800675/v_commune_2023.csv"
url_backup = "https://minio.lab.sspcloud.fr/lgaliana/data/python-ENSAE/cog_2023.csv"

try:
    cog_2023 = pd.read_csv(url_cog_2023)
except requests.exceptions.Timeout:
    cog_2023 = pd.read_csv(url_backup)

noms_communes = (
    cog_2023
    .loc[cog_2023["TYPECOM"] == "COM", ["COM", "LIBELLE"]]
    .rename(columns={"COM": "CODGEO", "LIBELLE": "LIBGEO"})
)

filosofi = (
    niveau_vie
```

```

.merge(taux_pauvrete, on="CODGE0", how="outer")
.merge(population, on="CODGE0", how="outer")
.merge(noms_communes, on="CODGE0", how="left")
[["CODGE0", "LIBGE0", "POPULATION", "NIVVIE_MEDIAN", "TAUX_PAUVRETE"]]
)

filosofi["POPULATION"] = filosofi["POPULATION"].astype(int)
filosofi["dep"] = filosofi["CODGE0"].str[:2]

```

```

import pandas as pd

url = "https://data.ademe.fr/data-fair/api/v1/datasets/igt-pouvoir-de-rechauffement-global/convert"
emissions = pd.read_csv(url)
emissions["dep"] = emissions["INSEE commune"].str[:2]

emissions.head(2)

```

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01

We will also need the following *packages*

To obtain reproducible results, you can set the seed of the pseudo-random number generator.

```
np.random.seed(123)
```

2 Descriptive statistics by group

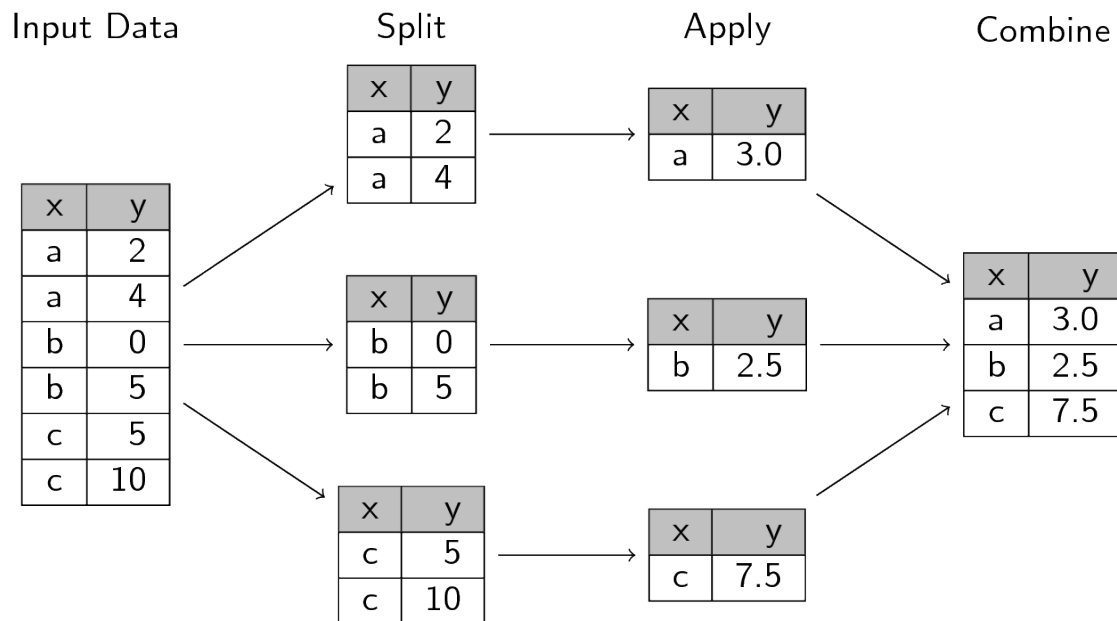
2.1 Principle

In the introductory chapter, we saw how to obtain an aggregated statistic easily with **Pandas**. However, it is common to have data with intermediate analysis strata that are relevant: geographical variables, membership in socio-demographic groups related to recorded characteristics, temporal period indicators, etc. To better understand the structure of the data, data scientists are often led to construct descriptive statistics on sub-groups present in the data.

For example, we previously constructed emission statistics at the national level. But what about the emission profiles of different departments? To answer this question, it will be useful to aggregate our data at the departmental level. This will give us different information from the initial dataset (municipal level) and the most aggregated level (national level).

In **SQL**, it is very simple to segment data to perform operations on coherent blocks and recollect results in the appropriate dimension. The underlying logic is that of *split-apply-combine*, which is adopted by data manipulation languages, including **pandas** which is no exception.

The following image, from [this site](#), well represents how the **split-apply-combine** approach works:

Figure 1: Split-apply-combine (Source: unlhcc.github.io)

In **Pandas**, we use **groupby** to segment the data according to one or more axes (this [tutorial](#) on the subject is particularly useful). All the aggregation operations (counting, averages, etc.) that we saw earlier can be applied by group.

Technically, this operation involves creating an association between labels (values of group variables) and observations. Using the **groupby** method does not trigger operations until a statistic is implemented; it simply creates a formal relationship between observations and groupings that will be used later:

```
filosofi.groupby('dep').__class__
```

```
pandas.api.typing.DataFrameGroupBy
```

As long as we do not call an action on a **DataFrame** by group, such as **head** or **display**, **pandas** performs no operations. This is called *lazy evaluation*. For example, the result of `df.groupby('dep')` is a transformation that has not yet been evaluated:

```
filosofi.groupby('dep')
```

```
<pandas.api.typing.DataFrameGroupBy object at 0x7f43bcd4d590>
```

2.2 Illustration 1: counting by group

To illustrate the application of this principle to counting, we can count the number of municipalities by department in 2023 (this statistic changes every year due to municipal mergers). For this, we simply take the reference of French municipalities from the official geographical code (COG) and count by department using **count**:

With this dataset, without resorting to group statistics, we can already know how many municipalities, departments, and regions we have in France, respectively:

```
communes = cog_2023.loc[cog_2023['TYPECOM']=="COM"]
communes.loc[:, ['COM', 'DEP', 'REG']].nunique()
```

Line 1

We restrict to the status “Commune” because this file also contains Insee codes for other statuses, such as the “Municipal Arrondissements” of Paris, Lyon, and Marseille.

```
COM      34945
DEP       101
REG        18
dtype: int64
```

Now, let’s look at the departments with the most municipalities. It is the same counting function where we play, this time, on the group from which the statistic is calculated.

Calculating this statistic is quite straightforward when you understand the principle of calculating statistics with **Pandas**:

```
communes = cog_2023.loc[cog_2023['TYPECOM']=="COM"]
communes.groupby('DEP').agg({'COM': 'nunique'})
```

COM	
DEP	
01	392
02	798
03	317
04	198
05	162
...	...
971	32
972	34
973	22
974	24
976	17

In SQL, we would use the following query:

```
SELECT dep, COUNT DISTINCT "COM" AS COM
FROM communes
GROUP BY dep
WHERE TYPECOM = 'COM';
```

The output is an indexed **Series**. This is not very convenient as we mentioned in the previous chapter. It is more practical to transform this object into a **DataFrame** with **reset_index**. Finally, with **sort_values**, we obtain the desired statistic:

```
(
    communes
    .groupby('DEP')
    .agg({'COM': 'nunique'})
    .reset_index()
```

```
.sort_values('COM', ascending = False)
)
```

	DEP	COM
62	62	890
1	02	798
80	80	772
57	57	725
76	76	708
...	...	...
96	971	32
99	974	24
98	973	22
100	976	17
75	75	1

2.3 Illustration 2: aggregates by group

To illustrate aggregates by group, we can use the Insee `filosofi` dataset and sum the `POPULATION` variable.

To calculate the total for the whole of France, we can do it in two ways:

```
filosofi['POPULATION'].sum()* 1e-6
```

```
np.float64(68.09428)
```

```
filosofi.agg({"POPULATION": "sum"}).div(1e6)
```

```
POPULATION    68.09428
dtype: float64
```

where the results are reported in millions of people. The logic is the same when doing group statistics, it's just a matter of replacing `filosofi` with `filosofi.groupby('dep')` to create a partitioned version of our dataset by department:

```
filosofi.groupby('dep')['POPULATION'].sum()
```

Line 1

With this approach, you need to pay attention to the order of operations: first, perform the `groupby` and then select the column of interest

```
dep
01    679344
02    523342
03    333298
04    168054
05    143467
```

```
...
92    1654712
93    1704316
94    1426929
95    1281653
97    1928465
Name: POPULATION, Length: 97, dtype: int64
```

```
filosofi.groupby('dep').agg({"POPULATION": "sum"})
```

	POPULATION
dep	
01	679344
02	523342
03	333298
04	168054
05	143467
...	...
92	1654712
93	1704316
94	1426929
95	1281653
97	1928465

The second approach is more practical because it directly gives a `Pandas DataFrame` and not an unnamed indexed series. From this, a few basic manipulations can suffice to have a shareable table on departmental demographics. However, this table would be somewhat rudimentary as we currently only have the department numbers. To get the names of the departments, we would need to use a second dataset and merge the common information between them (in this case, the department number): this is exactly the join principle we implemented in the [previous chapter](#).

3 Application

This application exercise uses the `Ademe` dataset named `emissions` previously discussed.

💡 Exercise 1: Group Aggregations

1. Calculate the total emissions of the “Residential” sector by department and compare the value to the most polluting department in this sector. Draw insights from the reality that this statistic reflects.
2. Calculate the total emissions for each sector in each department. For each department, calculate the proportion of total emissions coming from each sector.

Hint for this question

- “Group by” = `groupby`
- “Total emissions” = `agg({"**" : "sum"})`

In question 1, the result should be as follows:

	dep	Résidentiel	Résidentiel (% valeur max)
59	59	3.498347e+06	1.000000
75	75	1.934580e+06	0.552998
69	69	1.774653e+06	0.507283
62	62	1.738090e+06	0.496832
57	57	1.644192e+06	0.469991

This ranking may reflect demographics rather than the process we wish to measure. Without the addition of information on the population of each département to control for this factor, it is difficult to know whether there is a structural difference in behavior between the inhabitants of Nord (département 59) and Moselle (département 57).

At the end of question 2, let's take the share of emissions from agriculture and the tertiary sector in departmental emissions:

dep	Agriculture	Autres transports	Autres transports international	CO2 émissions hors total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	Part Agriculture	Part Autres transports	Part Autres transports international	Part CO2 émissions hors total	Part Déchets	Part Energie	Part Industrie hors-énergie	Part Résidentiel	Part Routier	Part Tertiaire	
23	1.43006e+06	566657501	0.000000	20116.604309	26558.838042	9752.578164	28626.245699	174077.313556	434767.608975	70773.260013	..	66.855172	0.215526	0.000000	8.944716	1.129566	0.410102	1.233163	5.710647	16.501132	5.009986
48	7.510929e+05	5697198142	0.000000	709105.948092	26811.391018	6865.546764	17883.285591	61053.916301	270618.408432	43661.121159	..	66.772488	0.461052	0.000000	5.737238	2.184764	0.490761	1.640564	4.938605	26.521761	3.532867
15	1.570206e+06	6261474910	0.000000	228415.910277	44814.873202	12138.432790	8324.449284	128353.601994	440832.993618	84364.615635	..	59.761414	0.520777	0.000730	1.799999	0.510115	5.308510	4.982805	17.212206	3.271036	
12	1.12225e+06	1376146978	0.000000	316420.919449	32612.611248	39105.647576	11097.496887	268862.644280	795613.916327	17021.964832	..	54.766487	0.391227	0.000004	8.466186	1.561495	0.966667	2.599427	6.891561	26.364560	4.205339
52	1.82660e+06	4399142452	0.000000	201732.701742	50956.68320	58651.612468	53468.498055	158218.060109	446405.991580	105662.674213	..	49.730704	0.223851	0.000000	9.771266	2.468261	0.896664	2.596235	7.664754	21.622845	5.118737

dep	Agriculture	Autres transports	Autres transports international	CO2 émissions hors total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	Part Agriculture	Part Autres transports	Part Autres transports international	Part CO2 émissions hors total	Part Déchets	Part Energie	Part Industrie hors-énergie	Part Résidentiel	Part Routier	Part Tertiaire	
75	0.000000	42214.820825	1.87700e+05	1.18657e+05	27358.781206	147963.117071	434014.468384	1.95638e+06	1.62358e+06	1.75163e+06	..	0.000000	0.627235	0.002730	17.630802	0.486495	2.194857	6.453018	28.743870	24.152808	19.781275
94	2209.429643	218992.353559	3.14629e+05	6.91405e+05	213619.061516	76361.230740	467189.089527	1.53689e+06	1.58943e+06	7.63650e+05	..	0.043081	4.167781	5.987888	13.138562	4.865530	1.452898	8.891867	25.445275	22.256195	14.531085
92	91.488184	12566.794939	2.10179e+02	1.06789e+02	264697.889711	242842.038012	786397.424067	1.86679e+06	1.59842e+06	8.36013e+05	..	0.001977	0.212030	0.000423	18.423930	4.563095	4.199841	12.197161	25.388352	26.877765	14.424724
99	2016.470962	57667.586124	1.16910e+06	7.20915e+05	202166.965778	10207.665783	432016.369996	1.53643e+06	1.39891e+06	8.40617e+05	..	0.001277	0.970766	17.611267	11.689538	4.052335	1.664458	4.937483	21.091146	12.397761	15.889109
83	151715.507862	21772.749176	2.85077e+04	5.79588e+05	253522.964895	47664.866369	139718.978613	5.93836e+05	1.56426e+06	5.61046e+05	..	3.527399	0.596209	0.603736	15.474887	5.629428	1.093778	5.248291	15.866766	45.208334	13.644031

These results are quite logical; rural departments have a larger share of their emissions from agriculture, while urban departments have higher emissions from the tertiary sector, which is related to the higher density of these areas.

With these statistics, we progress in understanding our dataset and, consequently, the nature of CO2 emissions in France. Descriptive statistics by group help us better grasp the spatial heterogeneity of our phenomenon.

However, we would remain limited in our ability to interpret these statistics without additional information: a département could look like a low emitter simply because it is sparsely populated. This is exactly the kind of limitation we resolved thanks to enriching our data with a join against the Filosofi data (see in particular the computation of a carbon footprint per capita in the previous chapter).

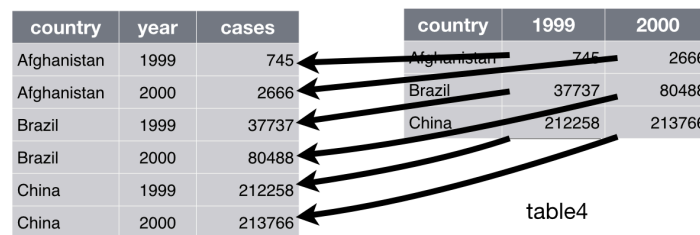
4 Restructuring datasets

4.1 Principle

When we have multiple pieces of information for the same individual or group, we generally find two types of data structures:

- **Wide** format: the data contains repeated observations for the same individual (or group) in different columns.
- **Long** format: the data contains repeated observations for the same individual in different rows, with a column distinguishing the observation levels.

An example of the distinction between the two can be taken from Hadley Wickham's reference book, *R for Data Science*:



country	year	cases
Afghanistan	1999	745
Afghanistan	2000	2666
Brazil	1999	37737
Brazil	2000	80488
China	1999	212258
China	2000	213766

country	1999	2000
Afghanistan	745	2666
Brazil	37737	80488
China	212258	213766

table4

Figure 2: Wide and Long Data Formats (Source: *R for Data Science*)

The following cheat sheet will help remember the functions to apply if needed:

Reshaping Data – Change the layout of a data set

 pd.melt(df) Gather columns into rows.	 df.pivot(columns='var', values='val') Spread rows into columns.
 pd.concat([df1, df2]) Append rows of DataFrames	 pd.concat([df1, df2], axis=1) Append columns of DataFrames

df.sort_values('mpg')
Order rows by values of a column (low to high).

df.sort_values('mpg', ascending=False)
Order rows by values of a column (high to low).

df.rename(columns = {'y': 'year'})
Rename the columns of a DataFrame

df.sort_index()
Sort the index of a DataFrame

df.reset_index()
Reset index of DataFrame to row numbers, moving index to columns.

df.drop(['Length', 'Height'], axis=1)
Drop columns from DataFrame

Switching from a *wide* format to a *long* format (or vice versa) can be extremely practical because certain functions are more suitable for one form of data than the other.

Generally, with **Python** as with **R**, **long formats are often preferable**. Wide formats are rather designed for spreadsheets like **Excel**, where we have a limited number of rows to create pivot tables from.

4.2 Application

The ADEME data, and the Insee data as well, are in the *wide* format. The next exercise illustrates the benefit of converting from *long* to *wide* before creating a plot with the **plot** method seen in the introductory chapter.

💡 Exercice 2: Restructuring Data: Wide to Long

1. Create a copy of the ADEME data by doing `df_wide = emissions.copy()`
2. Restructure the data into the *long* format to have emission data by sector while keeping the commune as the level of analysis (pay attention to other identifying variables).
3. Sum the emissions by sector and represent it graphically.
4. For each department, identify the most polluting sector.

5 Formatting descriptive statistics tables

A **Pandas** DataFrame is automatically formatted when viewed from a notebook as a minimally styled HTML table. This formatting is convenient for viewing data, a necessary task for data scientists, but it doesn't go much beyond that.

In an exploratory phase, it can be useful to have a more complete table, including minimal visualizations, to better understand the data. In the final phase of a project, when communicating about it, having an attractive visualization is advantageous. The outputs of notebooks are not a satisfactory solution for these needs and require the medium of the notebook, which can deter some users.

Fortunately, the young package **great_tables** allows for the creation of tables programmatically that rival tedious manual productions in **Excel** and are difficult to replicate. This package is a **Python** port of the **GT** package. **great_tables** builds *HTML* tables, offering great formatting richness and excellent integration with **Quarto**, the reproducible publishing tool developed by RStudio.

The following exercise will propose building a table with this package, step by step. It requires installing the **great_tables** package beforehand:

```
!pip install great_tables --quiet
```

To focus on table construction, the necessary data preparations are provided directly. We will start from the **emissions_merged** dataset, built in the previous chapter when computing the carbon footprint per capita, which looks like this:

To ensure you are able to complete the next exercise, here is the dataframe required for it.

```
emissions_totales_commune = emissions.sum(axis = 1, numeric_only = True)

emissions_merged = (
    emissions.assign(emissions = emissions_totales_commune)
    .reset_index()
    .merge(filosofi, left_on = "INSEE commune", right_on = "CODGEO")
)
emissions_merged = emissions_merged.loc[emissions_merged['POPULATION'] > 0]
emissions_merged['empreinte'] = emissions_merged['emissions']/emissions_merged['POPULATION']
emissions_merged['empreinte'] = emissions_merged['empreinte'].astype(float)
```

```
emissions_table = (
    emissions_merged
    .rename(columns={"dep_y": "dep", "POPULATION": "population", "NIVVIE_MEDIAN": "revenu"})
    .groupby("dep")
    .agg({"empreinte": "sum", "revenu": "median", "population": "sum"}) #pas vraiment le revenu
    .reset_index()
    .sort_values(by = "empreinte")
)
```

In this table, we will include horizontal bars, similar to the examples shown [here](#). This is done by directly including the *HTML* code in the DataFrame column.

```
def create_bar(prop_fill: float, max_width: int, height: int, color: str = "green") -> str:
    """Create divs to represent prop_fill as a bar."""
    width = round(max_width * prop_fill, 2)
    px_width = f"{width}px"
    return f"""\
<div style="width: {max_width}px; background-color: lightgrey;">\
```

```

<div style="height:{height}px;width:{px_width};background-color:{color};"></div>
</div>
"""

colors = {'empreinte': "green", 'revenu': "red", 'population': "blue"}

for variable in ['empreinte', 'revenu', 'population']:
    emissions_table[f'raw_perc_{variable}'] = emissions_table[variable]/
    emissions_table[variable].max()
    emissions_table[f'bar_{variable}'] = emissions_table[f'raw_perc_{variable}'].map(
        lambda x: create_bar(x, max_width=75, height=20, color = colors[variable])
    )

```

We keep only the 5 smallest carbon footprints and the five largest.

```

emissions_min = emissions_table.head(5).assign(grp = "5 départements les moins
pollueurs").reset_index(drop=True)
emissions_max = emissions_table.tail(5).assign(grp = "5 départements les plus
pollueurs").reset_index(drop=True)

emissions_table = pd.concat([
    emissions_min,
    emissions_max
])

```

Finally, to use some practical functions for selecting columns based on patterns, we will convert the data to the **Polars** format.

```

import polars as pl
emissions_table = pl.from_pandas(emissions_table)

```

💡 Exercise 5: A Beautiful Descriptive Statistics Table (Open Exercise)

Using this base table

```
GT(emissions_table, groupname_col="grp", rowname_col="dep")
```

construct a table in the style of the one below.

```

# Start from here
GT(emissions_table, groupname_col="grp", rowname_col="dep")

```

	empreinte	revenu	population	raw_perc_empreinte	bar_empreinte	raw_perc_revenu	bar_revenu	raw_perc_population	bar_population
5 départements les moins pollueurs									
92	142.01712816633915	36310.0	0.06576722880870661455	1.0	0.6326234356093262				
93	147.9911027308174	21820.0	0.07080916936246460305	0.6009363811622143	0.6515878553391433				
94	196.38841932390835	27940.0	0.1026738230470535274	0.7694849903607821	0.5455382727330075				
90	895.5043935192521	29170.0	0.104295486563765511	0.8033599559350041	0.05362177826799228				
83	922.1520421556788	26130.0	0.1050427191994636078	0.7196364637840815	0.4279293555866931				
5 départements les plus pollueurs									
52	13068.456519747897	24610.0	0.688451640180125334	0.6777747177086202	0.06435569183009097				
55	13651.074978199267	25340.0	0.870294207400370646	0.6978793720738089	0.06892781294026117				

	empreinte	revenu	population	parc_empreinte	bar_empreinte	bar_revenu	bar_population	bar_population
51	14401.966686862248	28180.0	5673874	783963581239	0.7760947397411182	0.21527315546702808		
21	15405.932865409246	27090.0	548426	737931792529	0.7460754613054255	0.2064890552389764		
77	18288.713383716273	28910.0	1468108	1.0	0.7961993941063068	0.5612816772982469		

The table you should have :

Carbon Footprint						
Initial descriptive statistics to refine						
Footprint			Median Income		Population	
Carbon Footprint	(%)*		Income	(%)*	Population	(%)*
5 départements les moins pollueurs						
92	14.20	0.8%	36.3K	100.0%	1.65M	63.3%
93	14.80	0.8%	21.8K	60.1%	1.70M	65.2%
94	19.64	1.1%	27.9K	76.9%	1.43M	54.6%
90	89.55	4.9%	29.2K	80.3%	140.25K	5.4%
83	92.22	5.0%	26.1K	72.0%	1.12M	42.8%
5 départements les plus pollueurs						
52	1,306.85	71.5%	24.6K	67.8%	168.33K	6.4%
55	1,365.11	74.6%	25.3K	69.8%	180.29K	6.9%
51	1,440.20	78.7%	28.2K	77.6%	563.08K	21.5%
21	1,540.59	84.2%	27.1K	74.6%	540.10K	20.6%
77	1,828.87	100.0%	28.9K	79.6%	1.47M	56.1%
*Note: The median income presented here is an approximation of the department's median income.						
Reading: The (%) columns presented above are scaled to the maximum value of the variable						
Source: Calculations based on Ademe data						

Thanks to this, we can already understand that our definition of the carbon footprint is certainly flawed. It seems unlikely that the inhabitants of the 77th department have a carbon footprint 500 times greater than that of intra-muros Paris. The main reason? We are not dealing with a concept of consumption emissions but production emissions, which penalizes industrial areas or areas with airports...

To learn more about constructing tables with `great_tables`, you can replicate this [exercise](#) on producing electoral tables that I proposed for an `R` course with `gt`, the equivalent of `great_tables` for `R`. We have now covered the main features of `Pandas` for manipulating and reshaping data. The [last chapter](#) of this part takes a step back to look at the limitations of `Pandas` syntax and discover the alternative ecosystems that exist to go further.

Informations additionnelles

i Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via `uv`. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```

[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette",
    "contextily=1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal=3.11.4",
    "graphviz=0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru=0.7.3",
    "markdown≥3.8",
    "nbcClient≥0.10.0",
    "nbformat≥5.10.4",
    "nltk≥3.9.1",
    "pandas≥3.0",
    "pip≥25.1.1",
    "plotly≥6.1.2",
    "plotnine≥0.15",
    "polars≥1.8.2",
    "pyarrow≥17.0.0",
    "pynsee≥0.1.8",
    "python-dotenv≥1.0.1",
    "python-frontmatter≥1.1.0",
    "pywaffle≥1.1.1",
    "requests≥2.32.3",
    "scikit-image≥0.24.0",
    "scikit-learn≥1.8.0",
    "scipy≥1.13.0",
    "seaborn≥0.13.2",
    "selenium<4.39.0",
    "spacy≥3.8.4",
    "webdriver-manager≥4.0.2",
    "wordcloud=1.9.3",
]

[tool.uv.sources]
cartiflette = { git = "https://github.com/inseeFrLab/cartiflette" }
gdal = [
    { index = "gdal-wheels", marker = "sys_platform == 'linux'", },
    { index = "geospatial_wheels", marker = "sys_platform == 'win32'", },
]

[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcDougall.github.io/geospatial-wheels-index/"

```

```
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean ≥ 4.0.1",
]
```

To use exactly the same environment (version of `Python` and `packages`), please refer to the [documentation](#) for `uv`.

Bibliography